

# Financial data for research: An advanced introduction to Refinitiv Workspace

From data discovery to reproducible access with Python

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Seminar for Doctoral Researchers

# Seminar overview

## Agenda

- 1 Refinitiv and the LSEG data ecosystem
- 2 Workspace interface and navigation
- 3 Companies, financial instruments, and identifiers
- 4 Available financial and non-financial information
- 5 Data Item Browser
- 6 CodeCreator and CodeBook
- 7 Connecting Workspace to Python

## By the end of the seminar

Participants will be able to:

- Navigate Refinitiv Workspace and locate information relevant to their research.
- Distinguish between companies, securities, and market identifiers.
- Identify appropriate data fields and understand their parameters.
- Build and execute data requests using Python.
- Adapt the provided GitHub examples to their own research projects.

# From Reuters to LSEG Workspace

The platform has evolved through several corporate and technological transformations.

| 1851                                                                                                 | 2008                                                                                           | 2018                                                                                    | 2021 onward                                                                                           |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Reuters</b><br>Paul Julius Reuter establishes a financial information and news service in London. | <b>Thomson Reuters</b><br>The Thomson Corporation and Reuters combine to form Thomson Reuters. | <b>Refinitiv</b><br>The Financial & Risk business of Thomson Reuters becomes Refinitiv. | <b>LSEG</b><br>LSEG acquires Refinitiv and integrates its data, analytics, and technology businesses. |

**Terminology used in this seminar:** Refinitiv refers to the data and analytics ecosystem familiar to many researchers. **LSEG Workspace** is the current desktop platform, while **Eikon** refers to its predecessor.

# The Workspace research ecosystem

Workspace brings together financial data, company information, news, and analytical tools within a single research environment.

## Company and reference data

- Company profiles and classifications
- Securities and market identifiers
- Ownership and corporate structures
- Officers, directors, and subsidiaries

## Fundamentals and estimates

- Financial statements and ratios
- Analyst estimates and recommendations
- ESG indicators and controversies
- Credit ratings and risk measures

## Market data

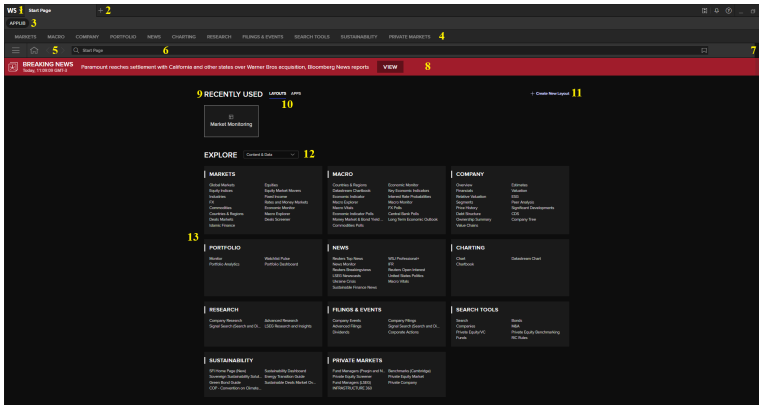
- Prices, returns, and trading volume
- Equity, debt, currency, and index data
- Current and historical observations
- Corporate actions and dividends

## News and analytical tools

- Reuters news and company events
- Charts and comparative analysis
- Screening and peer comparison
- Portfolio and industry analysis

**Access channels:** Workspace desktop and web applications, Microsoft Excel, CodeBook, and Python.

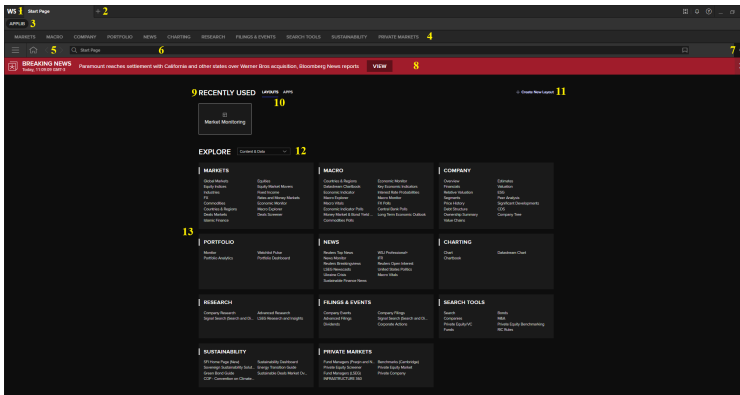
# Workspace interface and navigation



Main Workspace navigation elements. Interface may vary according to version and subscription.

- 1 **Workspace menu**  
Main Workspace access point.
- 2 **Workspace tab**  
Current open page or layout.
- 3 **App Library**  
Catalogue of Workspace apps.
- 4 **Content navigation**  
Markets, company, news, research, and tools.
- 5 **Main toolbar**  
Menu, home, and navigation controls.
- 6 **Search bar**  
Search companies, instruments, identifiers, and apps.
- 7 **Right-side controls**  
Layout and display options.
- 8 **Breaking news**  
Real-time news alert bar.

## Start page and content areas



**9 Recently used**  
Recently opened layouts and applications.

**10 Layouts / apps**  
Switch between layouts and individual apps.

**11 Create new layout**  
Build a customised working environment.

## 12 Explore filter

Filter visible content areas and tools.

**13 Content areas**  
Grouped resources by research domain.

Applications and content areas depend on user permissions and institutional subscription.

## How can we search in Workspace?

Workspace allows users to search by company name, market identifiers, and standardized financial or legal identifiers.

### Common search inputs

- **Name:** company or instrument name.  
Example: Amazon.com Inc
- **Ticker:** exchange-level trading symbol.  
Example: AMZN
- **RIC:** Refinitiv Instrument Code.  
Example: AMZN.O
- **ISIN:** international securities identifier.  
Example: US0231351067

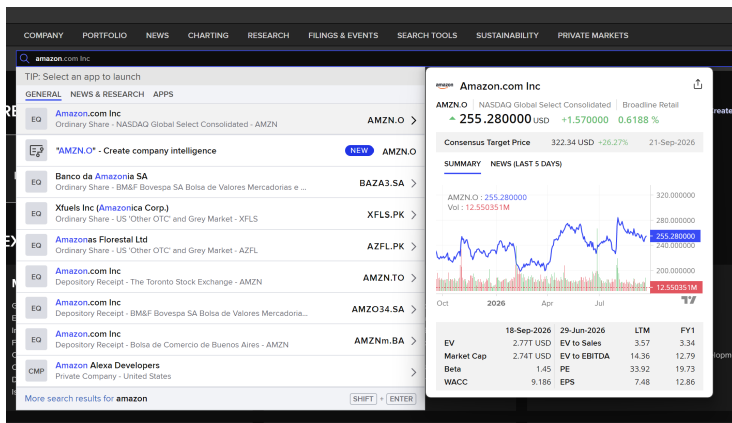
### Research-oriented identifiers

- **PermID:** persistent LSEG identifier for entities, instruments, and related objects.  
Useful for reproducible data work.
- **LEI:** Legal Entity Identifier.  
Identifies legal entities, not individual securities.
- **CUSIP / SEDOL:** local or market-specific identifiers.  
Common in equity and fixed-income datasets.

**Practical rule:** search broadly by name, but extract data using stable identifiers whenever possible.

# Live demo 1: Searching for a company

## Example: Amazon.com Inc



### What to look for

- 1 Search input**  
Enter a company name or identifier, such as Amazon or AMZN.O.
- 2 Search results**  
Identify companies, securities, and related applications.
- 3 Instrument identifier**  
AMZN.O identifies Amazon's Nasdaq-listed ordinary shares.
- 4 Information preview**  
Examine prices, charts, financial indicators, and news.

# Interpreting company search results

A single company may appear through different financial instruments and markets.

## What appears in the search results?

- **Company name:** Amazon.com Inc.
- **Primary listing:** Ordinary shares traded on Nasdaq.
- **Alternative instruments:** Depositary receipts and other securities.
- **Market identifiers:** Different RICs for different instruments.
- **Related applications:** Company intelligence and analytical tools.

## What should researchers check?

- **Company:** Is this the correct business entity?
- **Instrument:** Shares, depositary receipts, or bonds?
- **Market:** On which exchange is it traded?
- **Quotation:** Check price and trading currency.
- **Identifier:** Which RIC is appropriate for the analysis?

## Example: Amazon across different markets

**AMZN.O**  
Nasdaq (United States)

Ordinary shares

**AMZN.TO**  
Toronto (Canada)

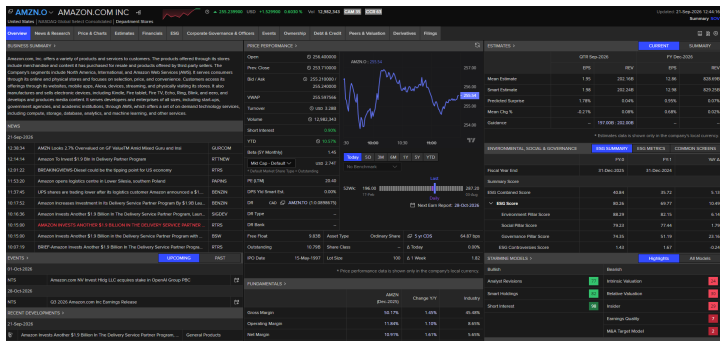
Depositary receipts

**AMZO34.SA**  
B3 (Brazil)

Brazilian depositary receipts

**Research takeaway:** Different instruments may have different prices, currencies, and market characteristics. Always record the exact identifier used in your data request.

# Amazon in Workspace: overview page



## What can we see here?

- Instrument header: company name, ticker, exchange, and market.
- Current market information: price, return, volume, and update time.
- Overview tab with business summary, news, events, and recent developments.
- Price performance panel with intraday movement and selected metrics.
- Snapshot panels with estimates, ESG indicators, and key fundamentals.

**Main idea:** This page gives a compact overview of the instrument and helps researchers quickly assess market data, firm information, analyst expectations, and sustainability indicators before moving to more detailed tabs.

# Key tabs for research

The top navigation tabs provide access to different types of data and analysis.

## Market and firm information

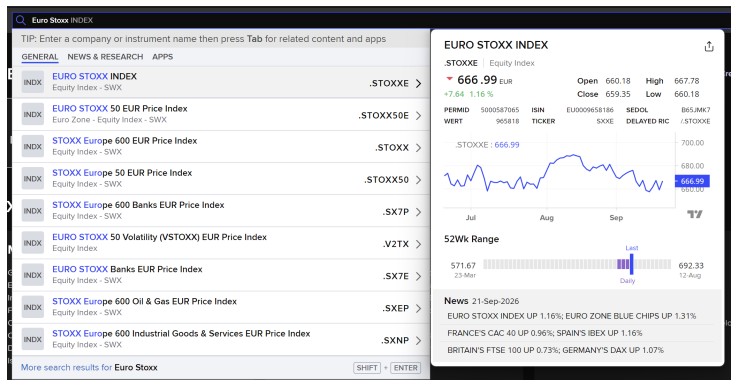
- **Overview:** quick summary of the company, news, and events.
- **Price & Charts:** historical prices, returns, and chart analysis.
- **Financials:** income statement, balance sheet, cash flow, and ratios.
- **Ownership:** institutional holdings, shareholders, and ownership structure.
- **Filings:** company reports and regulatory disclosures.

## Forward-looking and non-financial information

- **Estimates:** analyst forecasts for EPS, revenues, and target prices.
- **ESG:** environmental, social, and governance scores and metrics.
- **Corporate Governance & Officers:** executives, directors, and governance data.
- **Peers & Valuation:** comparable firms and relative valuation metrics.
- **Debt & Credit / Derivatives:** debt structure, credit risk, and related instruments.

Research examples: use **Financials** for accounting variables, **Estimates** for analyst expectations, **ESG** for sustainability research, and **Ownership** for studies on governance and institutional investors.

## Live demo 2: Searching for an index

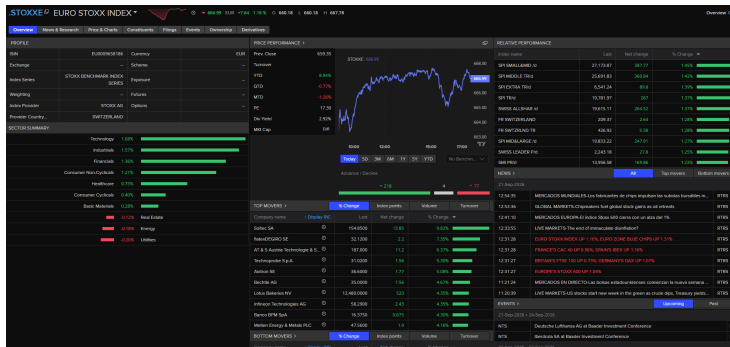


### Example: Euro Stoxx index

- Search by index name, ticker, RIC, or ISIN.
- Workspace returns several related index families and variants.
- Each result has a specific identifier, such as .STOXXE or .STOXX50E.
- Check the index type, currency, region, and methodology before using the data.
- The preview panel shows level, daily change, range, identifiers, chart, and news.

**Key research concept:** An index is not a company. It is a benchmark or constructed market indicator. Researchers should verify whether they need a broad index, a sector index, a price index, a total-return index, or a volatility index.

# Inside an index: similarities and differences



## Similarities with a company

- Overview and instrument identifiers.
- Historical charts and market performance.
- News, events, and related information.
- Access to additional data through specialised tabs.

## What is different?

- Index profile: Provider, currency, and index series.
- Constituents: Companies included in the index.
- Sector summary: Performance across economic sectors.
- Top movers: Constituents driving market movements.
- Relative performance: Comparison with other indices.

# Beyond companies and indices: what else can we search?

Workspace can also be used to search macroeconomic, country, currency, and interest-rate information.

## Search examples

- **Country information**  
Search by country name or ISO code.  
Example: Chile, CHL, USA.
- **Currencies and FX rates**  
Search by currency code or currency pair.  
Example: USD/CLP, EUR/USD, GBP/USD.
- **Equity indices**  
Search by index name or index identifier.  
Example: Euro Stoxx, S&P 500, IPSA.

## Rates and benchmarks

- **Government bond yields**  
Search sovereign curves or specific maturities.  
Example: 2-year, 5-year, or 10-year Treasury yields.
- **Risk-free rate proxies**  
Short-term government bills are often used as empirical proxies.  
Example: 3-month Treasury bill rate.
- **Reference rates**  
Useful for discounting, cost of capital, and macro-finance work.  
Example: SOFR, Euribor, central bank policy rates.

**Practical takeaway:** The search logic is similar across objects, but the interpretation changes. A company, an index, a country, a currency pair, and an interest-rate series represent different data objects. Always verify the identifier, currency, market, frequency, and unit before downloading data.

# Data Item Browser (DIB)

| FATERS                           |                                                              | Reset All | Find Data Item                     | Recently Used | All Items | Description                                                                                                                                                                                                                                                                                                                                                                                           | Parameters |
|----------------------------------|--------------------------------------------------------------|-----------|------------------------------------|---------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| Search All Filter Categories     |                                                              |           |                                    |               |           |                                                                                                                                                                                                                                                                                                                                                                                                       |            |
| ASSET CLASS                      |                                                              |           |                                    |               |           |                                                                                                                                                                                                                                                                                                                                                                                                       |            |
| Filter by Asset Class            |                                                              |           |                                    |               |           |                                                                                                                                                                                                                                                                                                                                                                                                       |            |
| CONTENT CLASSIFICATION (%)       |                                                              |           |                                    |               |           |                                                                                                                                                                                                                                                                                                                                                                                                       |            |
| Filter by Content Classification |                                                              |           |                                    |               |           |                                                                                                                                                                                                                                                                                                                                                                                                       |            |
| DATA TYPES                       |                                                              |           |                                    |               |           |                                                                                                                                                                                                                                                                                                                                                                                                       |            |
| Filter by Data Type              |                                                              |           |                                    |               |           |                                                                                                                                                                                                                                                                                                                                                                                                       |            |
|                                  | 451(K) Plan Expense                                          |           | TRF 401(K)PlanExp                  |               |           | 401(K) Plan Expense                                                                                                                                                                                                                                                                                                                                                                                   |            |
|                                  | 451(K) Plan Expense - Domestic                               |           | TRF 401(K)PlanExpDom               |               |           | TRF 401(K)PlanExp                                                                                                                                                                                                                                                                                                                                                                                     |            |
|                                  | 451(K) Plan Expense - Foreign                                |           | TRF 401(K)PlanExpFor               |               |           |                                                                                                                                                                                                                                                                                                                                                                                                       |            |
|                                  | 451(K) Plan Expense - Total                                  |           | TRF 401(K)PlanExpTot               |               |           |                                                                                                                                                                                                                                                                                                                                                                                                       |            |
|                                  | Above Market Rent/Losses - Account Amortization & Impairment |           | TRF AboveMktRentLossAccumImp       |               |           | Debit/Debit                                                                                                                                                                                                                                                                                                                                                                                           |            |
|                                  | Above Market Rent/Losses - Gross                             |           | TRF AboveMktRentLossGross          |               |           | 401(K) Plan Expense [401(K) Plan Expense] represents payments made to 401(K) defined contribution pension plans during the reporting period. The code reflects the sum of domestic and foreign pension plans. It is also used when a company does not specify whether a person plan is classified as domestic or foreign. Applicable to individuals. 401(K) Plan Expense [401(K) Plan Expense - Total |            |
|                                  | Above Market Rent/Losses - Net                               |           | TRF AboveMktRentLossNet            |               |           | [S401]                                                                                                                                                                                                                                                                                                                                                                                                |            |
|                                  | Acceptances Outstanding                                      |           | TRF AcceptOuts                     |               |           |                                                                                                                                                                                                                                                                                                                                                                                                       |            |
|                                  | Access Lines - Number - Total                                |           | TRF AccessLinesNumTot              |               |           |                                                                                                                                                                                                                                                                                                                                                                                                       |            |
|                                  | Accounting Change - Cash Flow - to Reconcile                 |           | TRF AccChgCf                       |               |           |                                                                                                                                                                                                                                                                                                                                                                                                       |            |
|                                  | Accounts & Notes Payable - Short Term, P&P Avg.              |           | TRF TruAccTTruNotesPbStPFAvg       |               |           |                                                                                                                                                                                                                                                                                                                                                                                                       |            |
|                                  | Accounts & Notes Payable - Short Term, P&P Avg, 5 Y Avg      |           | TRF TruAccTTruNotesPbStPFAvg5YAvg  |               |           |                                                                                                                                                                                                                                                                                                                                                                                                       |            |
|                                  | Accounts & Notes Payable - Short Term, P&P Diff              |           | TRF TruAccTTruNotesPbStPFDiff      |               |           |                                                                                                                                                                                                                                                                                                                                                                                                       |            |
|                                  | Accounts & Notes Payable - Short Term, WY Avg                |           | TRF TruAccTTruNotesPbStPFWAvg      |               |           |                                                                                                                                                                                                                                                                                                                                                                                                       |            |
|                                  | Accounts & Notes Payable - Short Term, WY Diff               |           | TRF TruAccTTruNotesPbStPWYDiff     |               |           |                                                                                                                                                                                                                                                                                                                                                                                                       |            |
|                                  | Accounts & Notes Receivable - Net - Total, 3 Y Avg           |           | TRF TruAccTTruNotesRcvNetTot3YAvg  |               |           |                                                                                                                                                                                                                                                                                                                                                                                                       |            |
|                                  | Accounts & Notes Receivable - Net - Total, 5 Y Avg           |           | TRF TruAccTTruNotesRcvNetTot5YAvg  |               |           |                                                                                                                                                                                                                                                                                                                                                                                                       |            |
|                                  | Accounts & Notes Receivable - Net - Total, 5 Y CAGR          |           | TRF TruAccTTruNotesRcvNetTot5YCAGR |               |           |                                                                                                                                                                                                                                                                                                                                                                                                       |            |
|                                  | Accounts & Notes Receivable - Net - Total, P&P Avg           |           | TRF TruAccTTruNotesRcvNetTotPFAvg  |               |           |                                                                                                                                                                                                                                                                                                                                                                                                       |            |
|                                  | Accounts & Notes Receivable - Net - Total, WY Avg            |           | TRF TruAccTTruNotesRcvNetTotFWAvg  |               |           |                                                                                                                                                                                                                                                                                                                                                                                                       |            |
|                                  | Accounts & Notes Receivable - Trade - Gross - Long Term      |           | TRF AccMktRcvTrdeGrossLT           |               |           |                                                                                                                                                                                                                                                                                                                                                                                                       |            |
|                                  | Accounts & Notes Receivable - Trade - Gross - Tot            |           | TRF AccMktRcvTrdeGrossTot          |               |           |                                                                                                                                                                                                                                                                                                                                                                                                       |            |
|                                  | Accounts & Notes Receivable - Trade - Net                    |           | TRF AccMktRcvTrdeNet               |               |           |                                                                                                                                                                                                                                                                                                                                                                                                       |            |
|                                  | Accounts & Notes Receivable - Trade - Net - Long Term        |           | TRF AccMktRcvTrdeNetLT             |               |           |                                                                                                                                                                                                                                                                                                                                                                                                       |            |
|                                  | Accounts Payable                                             |           | TRF AccPayble                      |               |           |                                                                                                                                                                                                                                                                                                                                                                                                       |            |
|                                  | Accounts Payable - Increase/Decrease - Cash Flow             |           | TRF AccCPACF                       |               |           |                                                                                                                                                                                                                                                                                                                                                                                                       |            |

## What is the DIB?

A tool to search, inspect, and select the data fields available in Workspace. It helps researchers identify the exact item code needed for data extraction.

## What should we look at?

- **Search bar:** find a variable by keyword.
- **Data Item Name:** human-readable field name.
- **Data Item Code:** field code used in extraction.
- **Description:** definition and interpretation.
- **Parameters:** optional settings or dimensions.
- **Filters:** narrow results by category.

# CodeCreator: from field selection to code

The screenshot displays the CodeCreator web application interface. At the top, there's a search bar for instruments, with 'AMZN.O' entered. Below this, the 'DATA ITEMS' section is active, showing a list of categories on the left and a list of fields on the right. The 'Total Assets' field is selected. The right panel provides a detailed description of 'Total Assets' (TR.F.TotAssets), stating it represents the total assets reported by a company and is the sum of Total Current Assets (XTCA) and Total Non-Current Assets (XNCA). Below the description, the 'Selected Items' section shows 'TR.F.TotAssets' has been added. The 'OUTPUT' section at the bottom shows the generated Python code for fetching this data using the Refinitiv Data Reader (rd) library. The code includes comments and uses the 'universe' and 'fields' parameters to specify the instrument and the field to be retrieved.

Instruments: **AMZN.O** Search Instruments

DATA ITEMS: **total assets**

Category: **Top Matches (20)**

- Real Time (55)
- Reference & Identifiers (1)
- Time Series Ratios (15)
- Price & Volume (10)
- Refinitiv I/B/E/S Estimates (6952)
- StarMine Models & Analytics (55)
- Green Revenue (6)
- Corporate Actions (2)
- Mutual Fund Holdings (2)

Fields:

- Total Assets**
- Other Assets - Total
- Total Current Assets
- Other Current Assets - Total
- Return on Average Total Assets - %, TTM
- Property Plant & Equipment - Net - Total
- Other Non-Current Assets - Total
- Intangible Assets - Total Pension
- Net Assets Recognized - Total Pension

**Add All** (20)

**Description**

### Total Assets

TR.F.TotAssets

Definition:

Total Assets [ATOT] represents the total assets reported by a company. In case the Total assets is not reported, it will be the sum of the following items: • Total Current Assets [XTCA] • Total Non-Current Assets [XNCA] Applicable to all industries. Category: Other Assets (Current & Non-Current)

Parameters & Quick Functions

Selected Items: **TR.F.TotAssets** Drag to Reorder [Clear All](#)

OUTPUT: **Python**

```
import refinitiv.data as rd
rd.open_session()
df = rd.get_data(
    universe = ['AMZN.O'],
    fields = ['TR.F.TotAssets']
)
display(df)
```

[Copy to Clipboard](#) [Open in CodeBook](#)

Main idea: CodeCreator reduces the distance between “I know what data I need” and “I know how to request it using code”.

# How CodeCreator supports reproducible data access

## Workflow

- 1 Select the instrument.  
Example: AMZN.O
- 2 Search and select the required field.  
Example: TR.F.TotAssets
- 3 Review the field definition.
- 4 Define parameters if needed.
- 5 Generate code for Python, Excel, or CodeBook.

## Why it is useful

- Connects field discovery with data extraction.
- Reduces coding errors in variable names.
- Shows the exact field code required.
- Helps move from Workspace to Python.
- Makes queries easier to document and reproduce.

**Research takeaway:** Use CodeCreator when you already know the instrument and want to translate a data need into a reproducible query.

# Working with LSEG data using Python

Two approaches to accessing financial data programmatically.

## CodeBook: Integrated environment

- Cloud-hosted Python development environment integrated into LSEG Workspace.
- Provides Jupyter notebooks and pre-installed Python libraries.
- Allows users to retrieve, analyse, and visualise financial data.
- No local Python installation is required.

**Workflow:** Workspace → CodeCreator → CodeBook → Data analysis.

## Python: External environment

- Connect to LSEG data using Python APIs from your own computer.
- Work with Jupyter Notebook, VS Code, or other Python environments.
- Integrate data extraction into your own research projects.
- Requires local configuration, authentication, and authorised data access.

**Workflow:** Python environment → LSEG API → Data extraction and analysis.

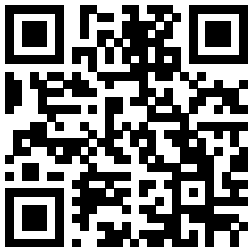
## Additional resource: Refinitiv with Python

For researchers interested in using Python outside CodeBook, the following GitHub repository provides practical examples and notebooks for accessing and analysing Refinitiv data.

[Refinitiv with Python – GitHub Repository](#)

# Thank you!

Thank you for participating in this seminar.



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